

Management of Shock

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 Adapted from: [Management of Shock](#)

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Learning Objectives (4)

Versions

After completing this brick, you will be able to:

- 1 Describe the hemodynamic findings for each type of shock.
- 2 Explain the use of fluid resuscitation in shock.
- 3 Justify the use of pharmacological agents in shock treatment
- 4 Outline the treatment plan for each type of shock



CASE CONNECTION

LT is brought to the emergency department by his son. “My dad lives with me. For the past few days, he has been confused, tired, and coughing up green stuff. This morning he said he wanted to get ready for his wedding, and Mom died 15 years ago.” On exam, LT is lethargic, and his vitals are a blood pressure of 80/62

mm Hg, heart rate is 116 beats/min, respiratory rate is 30 breaths/min, and temperature is 101.3°F. On physical exam, he has crackles at the right lower lung base, and his skin is cool. You start IV fluids, and labs and imaging are ordered. LT's white blood cell count and serum creatinine are markedly elevated. His chest x-ray demonstrates right lower lobe pneumonia. After receiving 2 L of IV fluids in the emergency department, LT becomes more lethargic, and his blood pressure does not improve.

What is causing LT's hemodynamic condition? What medications should be given next? Consider your answers as you read, and we'll revisit LT at the end of the brick.

[GO TO CONCLUSION](#) ↓

What Is Shock? (LO#1)

Shock occurs when blood flow to the body's tissues decreases below their oxygenation needs, which is most easily quantified by hypotension.

Clinical Significance

Shock is defined as a state of systemic hypoperfusion of tissues, resulting in widespread cellular and tissue hypoxia. This shortage is not immediately fatal, but it can rapidly progress to multiorgan system failure and death if not corrected quickly. This brick will focus on how we stabilize the patient while looking for the underlying cause of shock (eg, infection, trauma, or heart failure).

Types of Shock

The five types of shock are:

- **Distributive shock** arises from systemic vasodilation, which might be triggered by conditions such as sepsis or anaphylaxis.
- **Cardiogenic shock** can occur from a condition such as heart failure or myocardial infarction that leads to low stroke volume (SV) and low cardiac output (CO).
- **Hypovolemic shock** is caused by low intravascular volume that leads to low preload and low CO and can be caused by hemorrhage, GI blood loss, or severe dehydration.
- **Neurogenic shock** is caused by the loss of sympathetic output from the central nervous system (CNS), leading to a global loss of arterial tone, cardiac output, heart rate, and preload (venous tone); neurogenic shock can follow head or spinal cord trauma.
- **Obstructive shock** is due to an obstruction in blood flow through the heart. There are two types. In the pulmonary type, a pulmonary embolus causes increased resistance to flow within the pulmonary circulation and prevents blood from reaching the left side of the heart. In the structural type, there is global obstruction of flow through the heart caused by cardiac tamponade or tension pneumothorax.

ACTIVE LEARNING QUESTION

Which type of shock is caused by systemic vasodilation?

- A. Distributive shock
- B. Cardiogenic shock
- C. Hypovolemic shock

- D. Neurogenic shock
- E. Obstructive shock

Explanation (requires correct answer)

The hemodynamic findings of the different types of shock are shown in **Table 1**. The precipitating event is shown with a P, and a unique feature of a particular type of shock is shown with an asterisk.

In the previous session, we discussed preload, afterload, cardiac output (CO), and mean arterial pressure (MAP) as the primary physiologic determinants of perfusion. In clinical practice, particularly in critically ill patients, additional hemodynamic measurements help us better classify shock and guide therapy. These include **right atrial pressure (RAP)**, **pulmonary capillary wedge pressure (PCWP)**, **cardiac index (CI)**, and **mixed venous oxygen saturation (SvO₂)**. Together, these measurements help localize where in the circulatory system the problem is occurring — venous return, ventricular filling, pump function, or tissue oxygen utilization.

Table 1 Hemodynamic findings of shock

	Mean arterial blood pressure (MAP)	End-diastolic volume (preload)	Systemic vascular resistance	Cardiac output	Heart rate
Distributive	↓	↓	↓P	↑	↑
Cardiogenic	↓	↑	↑	↓P	↑
Hypovolemic	↓	↓P	↑	↓	↑
Neurogenic	↓P	↓P	↓P	↓P	*Normal or ↓
Obstructive: pulmonary	↓	*↑ (RV) ↓ (LV)	↑	↓P	↑
Obstructive: mechanical	↓	↓	↑	↓P	↑

Right Atrial Pressure (RAP): Right atrial pressure reflects the pressure in the right atrium and is often used as a surrogate for **right ventricular preload**. Clinically, RAP is equivalent to central venous pressure (CVP) when measured via a central venous catheter. Normal RAP ranges from **2–8 mmHg**.

An elevated RAP suggests impaired right-sided emptying or increased venous return pressure. This can be seen in: Cardiogenic shock (right ventricular failure) Obstructive shock (e.g., pulmonary embolism, cardiac tamponade) Volume overload states. A low RAP suggests decreased venous return, as seen in hypovolemic shock.



CLINICAL REASONING

RAP helps differentiate cardiogenic and obstructive shock from hypovolemic shock.

Pulmonary Capillary Wedge Pressure (PCWP) Pulmonary capillary wedge pressure estimates **left atrial pressure** and therefore reflects **left ventricular preload**. It is obtained using a pulmonary artery catheter, which measures pressure transmitted backward from the left side of the heart through the pulmonary circulation. Normal PCWP ranges from approximately **6–12 mmHg**. Low PCWP suggests reduced preload (e.g., hypovolemia). Elevated PCWP indicates increased left-sided filling pressures, as seen in cardiogenic shock, where blood backs up into the pulmonary circulation due to impaired ventricular pumping.



CLINICAL REASONING

PCWP, like RAP helps differentiate cardiogenic and obstructive shock from hypovolemic shock.

Cardiac Index (CI) Cardiac index adjusts cardiac output for body surface area

$$CI = CO/BSA$$

Normal CI is approximately **2.5–4.0 L/min/m²**. A reduced CI indicates inadequate forward flow relative to patient size and is seen in cardiogenic, hypovolemic, and obstructive shock. In early distributive (septic) shock, CI may be normal or elevated due to decreased systemic vascular resistance and

compensatory hyperdynamic circulation. (hyperdynamic = decrease in SVR due to endotoxin release leads to increase in CO, however, in late septic shock SVR may increase)

Mixed Venous Oxygen Saturation (SvO₂) Mixed venous oxygen saturation reflects the percentage of oxygen remaining in blood returning to the heart after tissue extraction. It represents the balance between oxygen delivery (DO₂) and oxygen consumption (VO₂). Normal SvO₂ ranges from **60–80%**. Low SvO₂ indicates that tissues are extracting more oxygen because delivery is inadequate (low cardiac output states such as cardiogenic or hypovolemic shock). Elevated SvO₂ may be seen in early septic shock due to impaired oxygen extraction at the tissue level.

Placement of a **right heart (Swan-Ganz) catheter (Figure 1)** can determine right atrial and left atrial pressure (PCWP), determine cardiac output, and allow precise calculation of SVR in unclear cases. Because the catheter is now recognized to have risks and does not improve mortality, it is used less frequently than before.

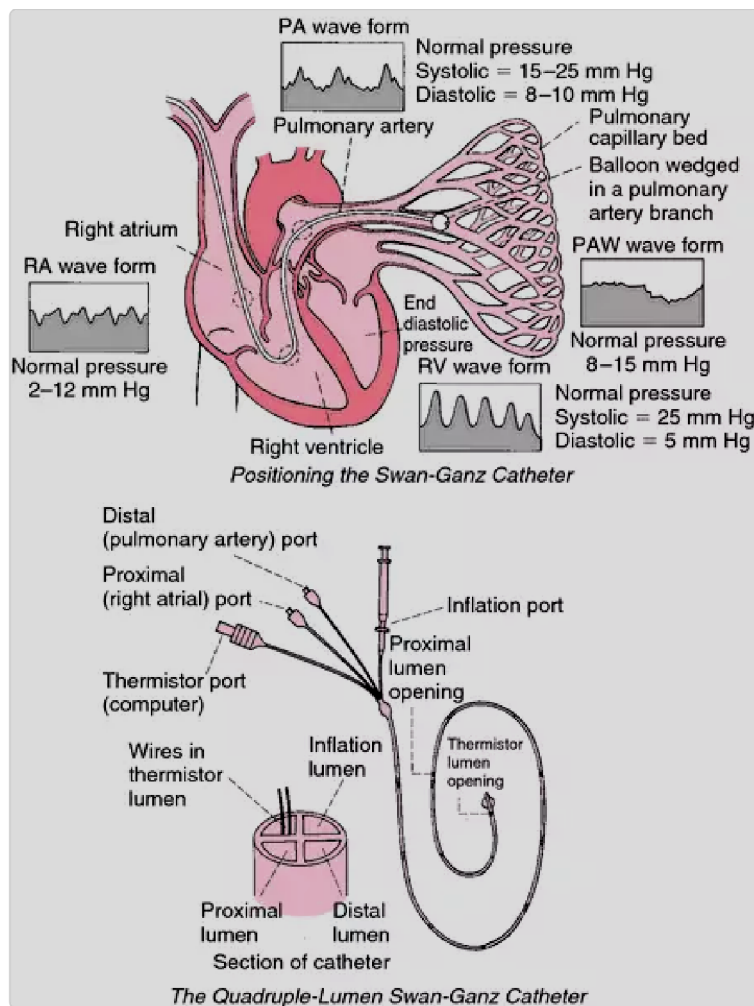


Figure 1: Swan-Ganz Catheter

Why Do We Initially Give Intravenous Fluids to Most Shock Patients? (LO#2)

When faced with a patient with hypotension, the first step is to stabilize blood flow and organ perfusion. To do this, give the patient a **bolus of IV fluids** (eg, lactated Ringer solution or normal saline), **unless** the patient has evidence of pulmonary edema (more on that in a minute).

Administering IV fluids is the most effective treatment of **hypovolemic shock**. Here, the primary insult is low cardiac preload. The lower preload causes reductions in stroke volume and cardiac output, which, in turn, cause low blood pressure. Fluid therapy increases preload, raising cardiac output.

The effect is most marked in those with hypovolemic shock but may be useful as well in those with mild distributive shock. Blood can be used instead of IV fluids but is generally reserved for those with overt hemorrhage.

Giving IV fluids carries little risk, which is why IV fluids are used in most hypotensive patients, even before the precise cause of low blood pressure or shock is identified. In fact, the patient's response to saline is often diagnostic: if they respond promptly, you are probably dealing with hypovolemic shock and not one of the other types.

When should you **not** give IV fluids? Patients should not receive IV fluids if they have pulmonary edema, diagnosed hypoxia, and interstitial edema on chest x-ray (Figure 1).

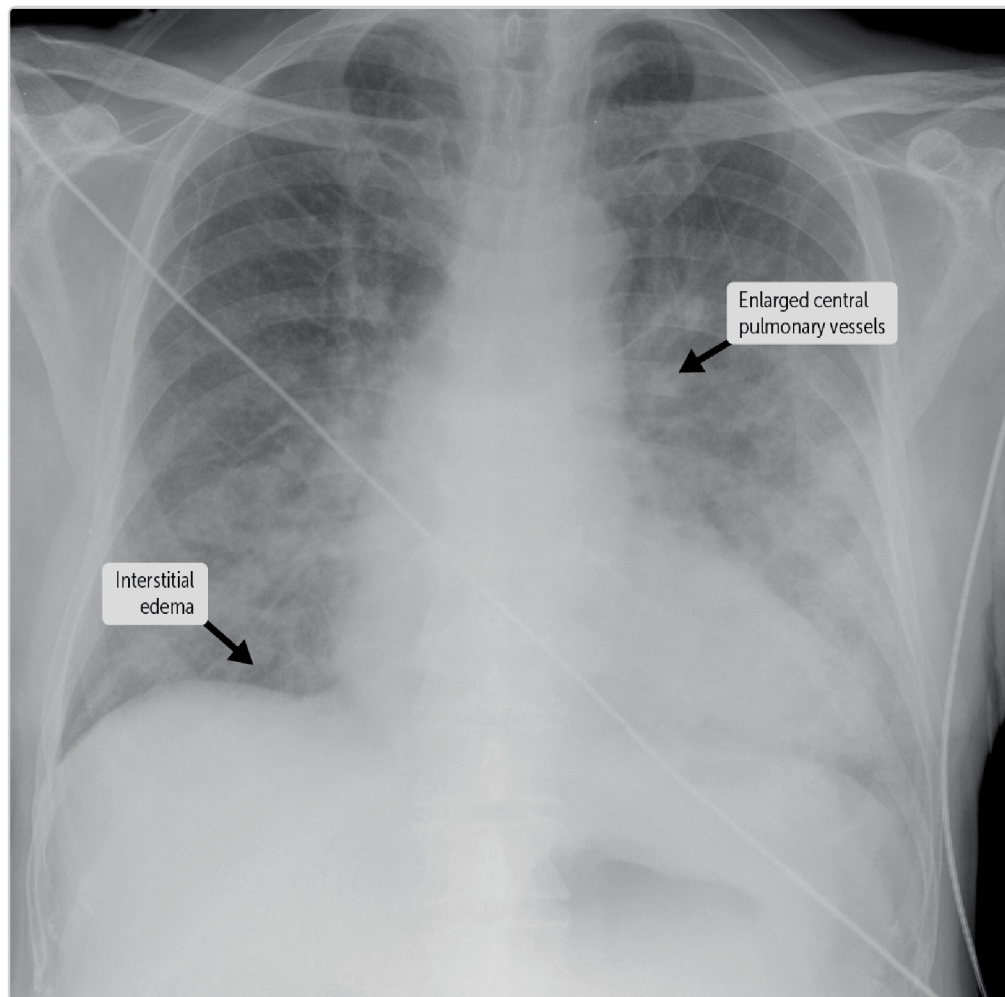


Figure 1 Pulmonary edema

They also usually have **pulmonary crackles** 🕒 on chest exam. In patients with pulmonary edema, IV fluids may worsen the hypoxia and potentially cause respiratory failure.

hypovolemic shock.

preload, and a positive response suggests an etiology of

Administering IV fluid serves two purposes: it increases a patient's

How Do We Manage Distributive Shock With Vasopressors? (LO#3)

As you see in Table 1, the different forms of shock have different physiologic patterns. Many of the therapies for shock are directed specifically against one of these abnormalities. We will now work through some of these, beginning with reduced systemic vascular resistance (SVR), one of the characteristics of distributive shock.

Distributive shock is caused by arteriolar vasodilation, which reduces SVR. Therefore, **vasopressor drugs** (sometimes called “pressors”), which cause constriction of the arterioles, are an effective treatment. The following is a

list of receptors and the pressor drugs that activate them to achieve vasoconstriction:

- α_1 -Adrenergic, activated by norepinephrine, epinephrine, or phenylephrine
- Vasopressin V_{1A} , activated by vasopressin
- Dopaminergic, activated by dopamine

Norepinephrine

Although the use of these pressor drugs varies and can be customized for an individual patient, norepinephrine is the drug of choice for most patients with distributive shock, as has been shown in survival studies. It acts primarily at the α_1 adrenergic receptor, secondarily at the β_1 , and β_2 receptors. The drug causes vasoconstriction and tachycardia (via β_1 stimulation).

An adverse effect is tachyarrhythmias, although the risk is lower than that of stronger β_1 agonists. Like the other pressor drugs, norepinephrine should only be used when absolutely necessary in pregnant patients, as the drugs may cause early pregnancy termination.

Other Vasopressors

While norepinephrine is the drug of choice for distributive shock, the drugs below may also be useful if the blood pressure remains low despite norepinephrine:

- **Epinephrine** has the most wide-ranging adrenergic affects and acts strongly at the α_1 , β_1 , and β_2 receptors. The overall effect is vasoconstriction, tachycardia, and positive inotropy. Epinephrine

frequently causes tachyarrhythmias and constriction of gut vessels, which is one reason why this drug is preferred less than norepinephrine for most patients.

- **Phenylephrine** specifically activates α_1 receptors. Therefore, it causes only vasoconstriction. Unlike the other agents that have been described, it can cause occasional reflex bradycardia: the drug-induced increase in blood pressure is detected by baroreceptors, which then trigger a reflexive parasympathetic signal to slow the heart rate.
- **Vasopressin**, also known as antidiuretic hormone (ADH), has many functions in the body, including vasoconstriction of smooth muscle cells. As a pressor drug, it acts through vasopressin-specific V_{1A} receptors to constrict the vascular smooth muscle cells. Vasopressin has no direct effect on heart rate because vasopressin receptors are not located on pacemaker cells in the heart.
- **Dopamine** is considered both a vasopressor and a weak inotrope (stimulator of cardiac contractility, see below). It acts not only through α - and β -adrenergic receptors, but also through **dopaminergic-specific receptors**. The most pharmacologically relevant dopamine-specific effect is to induce vasodilation in the renal circulation. The degree of activity on each type of receptor is dose-dependent: low doses are primarily dopaminergic, and increasing doses produce inotropic effects via β_1 stimulation, then systemic vasoconstriction α_1 stimulation. Side effects include tachycardia and tachyarrhythmias, as is the case with most drugs with inotropic effects.



CLINICAL CORRELATION

Because acute kidney injury is so common in shock-states, the theoretical benefit of low-dose dopamine is to preserve blood flow to the kidneys that sustains renal function. However, there has been no proven renal-protective benefit in the clinical setting, and increasing renal blood flow may lead to diuretic effect that can cause significant harm in a hypotensive patient.

These drugs can be used in combination, and it is common to mix drugs with different receptor targets (eg, use norepinephrine with vasopressin). [Table 2](#) summarizes the effects of the vasopressor drugs on the various receptors; the number of plus signs indicates relative receptor affinity. (**This is a must know table !!**)

Table 2 Vasopressor receptor targets

Drug	α_1	β_1	β_2	V _{1A}	Dopaminergic
Norepinephrine	++	+	+	None	None
Phenylephrine	+++	None	None	None	None
Epinephrine	+++	++	++	None	None
Vasopressin	None	None	None	+++	None
Dopamine	+	+	+	None	++

Phenylephrine selectively stimulates α_1 receptors.

The other class of medications used in treatment of Shock are **inotropes**.
The

Inotropes need to be used carefully in the setting of myocardial infarction (MI) or chest pain. This is because both inotropes so strongly stimulate the heart that the energy expended may exceed oxygen supply in patients with coronary ischemia or MI. Additionally, inotropic agents also tend to be chronotropic agents, increasing the risk of tachyarrhythmias that can be treatment-limiting and life-threatening.

- Dobutamine strongly stimulates β_1 receptors, causing an **increase in cardiac contractility and heart rate** and, therefore, cardiac output. By weakly stimulating β_2 receptors, dobutamine also causes mild arteriolar vasodilation, reducing SVR and afterload; hence, it is often helpful in treating heart failure. It has minimal effect on α_1 receptors, so causes little or no vasoconstriction. Dobutamine can raise the heart rate substantially and can also precipitate ventricular and atrial arrhythmias through its β_1 stimulation. Because it causes vasodilation, it can sometimes lower blood pressure.
- The inotrope **milrinone** works differently; it inhibits phosphodiesterase 3 ([PDE3] see [Figure 2](#)). PDE3 lies downstream, inside smooth muscle cells and on the β_1 pathway; PDE3 degrades cyclic adenosine monophosphate (cAMP) into AMP, which leads to a decrease in

intracellular Ca^{2+} and, therefore, a decrease in contractility. By inhibiting PDE3, milrinone decreases cAMP degradation and thus increases myocyte intracellular Ca^{2+} and cardiac contractility. This has the effect of **increasing stroke volume and cardiac output**. By inhibiting PDE3, milrinone also causes vasodilation throughout the vasculature. Milrinone’s combined effect of increasing contractility and decreasing afterload makes it great for treating cardiogenic shock.

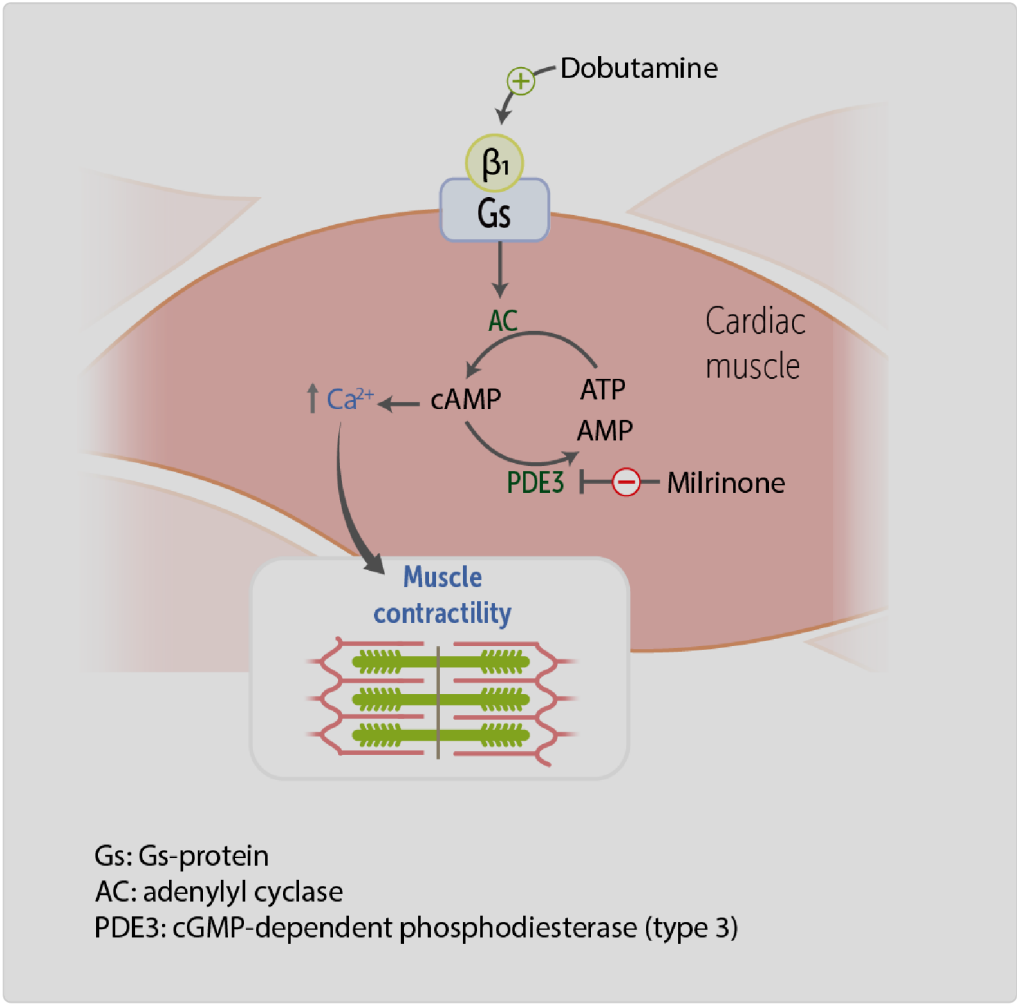


Figure 2 Dobutamine and milrinone signaling pathways

Table 3 compares the two main inotropic drugs.

Table 3 Comparison of dobutamine and milrinone

Drug	Contractility	Cardiac output	BP	HR	SVR	Mechanism
Dobutamine	↑↑↑	↑↑↑	Variable	↑↑↑	↓	$\beta_1 > \beta_2 \gg \alpha_1$
Milrinone	↑↑↑	↑↑↑	Variable	↑↑	↓↓	cAMP

cardiac contractility (inotropy).

Milrinone inhibits PDE3, leading to vasodilation and increased

Mechanical Devices

Some patients with severe cardiogenic shock do not respond to drugs alone and require placement of a **mechanical assist device** to support them until their hearts recover. None of the devices described below have demonstrated clear improvements in patient outcomes, including mortality. These devices should not be used in patients with sepsis, bleeding, or severe aortic disease:

- **Intra-aortic balloon pump (IABP):** This is a catheter-inserted pump that lies in the proximal aorta and coordinates with the ECG to assist the

heart in pumping blood forward (Figure 3).

- **Percutaneous left ventricle assist devices:** These require surgical placement and come with a higher risk than IABP. They can be used to support the left ventricle, even when a thrombus is present in-situ, and can also provide support if right ventricular failure occurs. In addition, these devices can be used in the presence of a prosthetic aortic valve or aortic stenosis.
- **Extracorporeal membrane oxygenation (ECMO):** This provides complete cardiopulmonary support for patients with both cardiac and respiratory failure. It involves pumping the patient's blood into a heart-lung machine for oxygenation and removal of carbon dioxide. Full hemodynamic support can be provided by this method. There are currently no specific indications to define when to start ECMO in patients with cardiogenic shock.

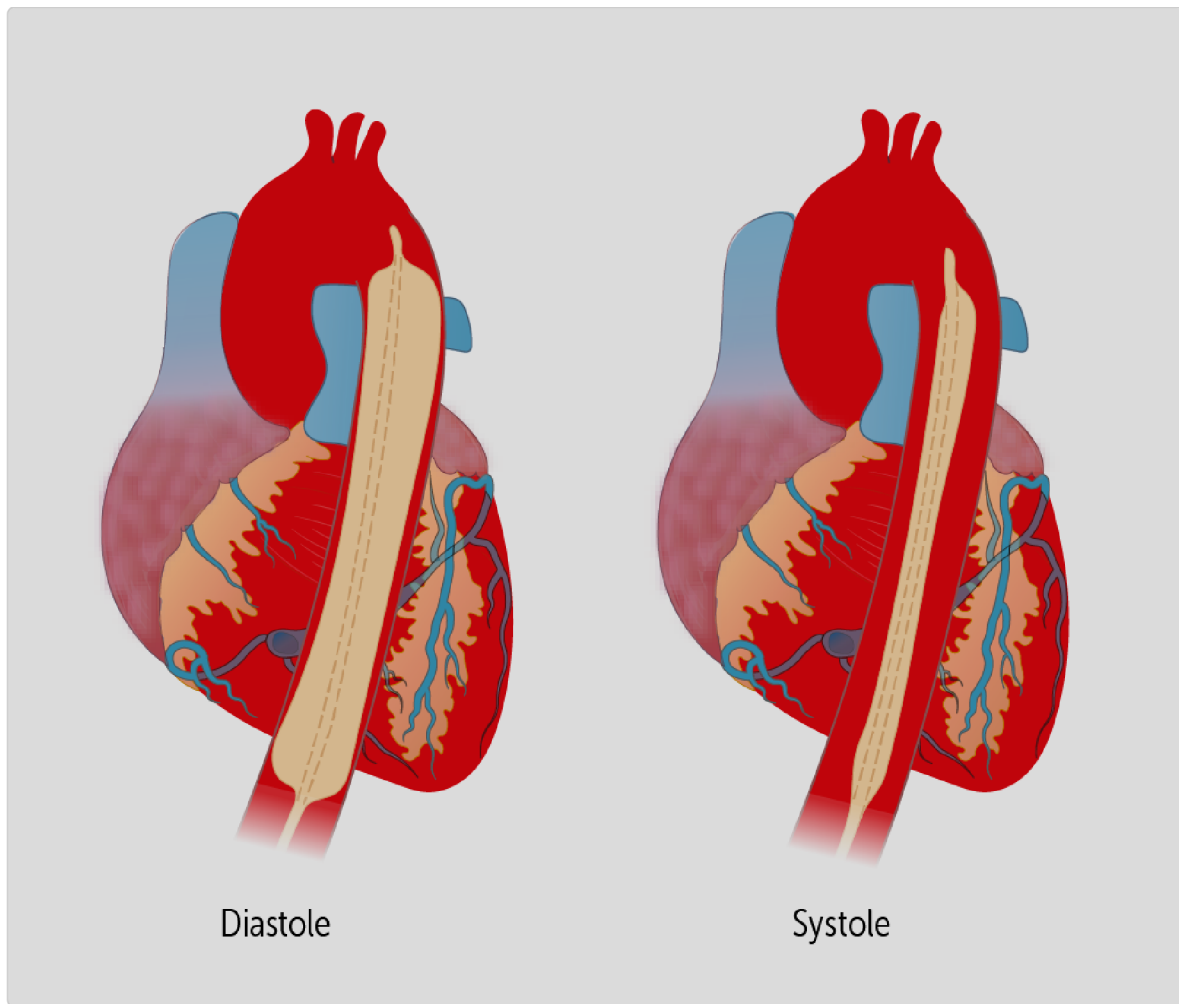


Figure 3 Intra-aortic balloon pump

How Are Other Types of Shock Treated? (LO#4)

The other types of shock have varied treatments.

Neurogenic shock is generally supported with IV fluids or, occasionally, pressor drugs until the autonomic nervous system recovers.

Cardiogenic shock: Inotropes are used to treat cardiogenic shock. The two inotropes used in cardiogenic shock are dobutamine and milrinone. As discussed, many adrenergic vasopressors also have strong inotropic activity, most notably epinephrine and dopamine.



CLINICAL CORRELATION

Early assessment of cardiac function is critical in shock evaluation. While formal echocardiography requires a technician, equipment transport, image acquisition, and delayed cardiologist interpretation, Point Of Care UltraSound (POCUS) allows you to assess cardiac function in real-time at the bedside. POCUS reveals the heart's systolic function, measured as ejection fraction (EF). A newly reduced EF in the setting of shock suggests cardiogenic etiology—though this determination requires knowledge of the patient's baseline function. Normal or supranormal EF points toward distributive, hypovolemic, or obstructive shock.

The **pulmonary type of obstructive shock** (from a pulmonary embolus) is treated initially with IV fluids (and pressor drugs if needed), then definitively by dissolving the embolus, often with thrombolytic drugs like tissue plasminogen activators.

The **structural causes of obstructive shock** are treated initially with IV fluids, then with surgical techniques that correct the obstruction:

- Cardiac tamponade is treated by withdrawing pericardial fluid (pericardiocentesis).
- Tension pneumothorax is treated with a thoracostomy tube placed to alleviate the high pressure in the pleural space.

How do we manage shock?

Table 4 summarizes the principles of shock management.

Table 4 Management of shock

Type of shock	Treatments
Distributive	IV fluids Pressor drugs: norepinephrine, epinephrine, phenylephrine, vasopressin, dopamine
Cardiogenic	Inotropic drugs: milrinone, dobutamine
Hypovolemic	IV fluids (or blood in acute hemorrhage)
Neurogenic	IV fluids Pressor drugs
Pulmonary obstructive (pulmonary embolism)	IV fluids Thrombolytic drugs
Structural obstructive: cardiac tamponade	IV fluids Pericardiocentesis
Structural obstructive: tension pneumothorax	IV fluids Thoracostomy

As you can see in [table 4](#), the treatment options are IV fluids, vasopressors, and inotropes.

Treatment: Hypovolemic Shock

- Crystalloids first: Normal saline or lactated Ringer's (LR is slightly preferred)
- Bolus strategy: Rapid IV boluses (500-1000mL), reassess, repeat as needed
- Blood products if bleeding: If hemorrhagic shock, transition to packed red blood cells (pRBCs) early.
- Monitor hemodynamic and organ perfusion response

Treatment: Distributive Shock

- Septic Shock (most common): Fluids first: 30mL/kg crystalloid bolus within 3 hours (well-established initial resuscitation)
- Antibiotics: Administered as soon as possible
- Vasopressors: Norepinephrine is first-line when fluids aren't enough
- Source control: Find and fix the infection (drain abscess, remove infected catheter, etc.)

Treatment: Anaphylactic Shock

- Epinephrine: first-line (EpiPen) to reverse vasodilation (α -1 mediated vasoconstriction will reverse) and bronchospasm (β 2 mediated bronchodilation will reverse)
- Fluids: Aggressive IV crystalloids
- Adjuncts: H1/H2 blockers, corticosteroids, albuterol if bronchospasm

hypotension.

For this type of shock, especially when these patients have severe
Thrombolytic drugs (eg, tissue plasminogen activator) can be used



Thinking back to LT, what is causing his hemodynamic condition? What medications should be administered next?

LT has pneumonia and sepsis. Byproducts of sepsis lead to arterial vasodilation, low SVR, and low blood pressure. In addition to giving LT broad-spectrum antibiotics to treat his pneumonia, you give him vasopressors to increase his SVR and blood pressure. After LT is admitted to the ICU, he improves over the course of 3 days. After those 3 days, he is transferred to a general medical unit and then discharged to home.

Summary

- Shock is a state of systemic hypoperfusion resulting in tissue hypoxia, potentially leading to multiorgan failure and death if untreated.
- There are five types of shock: distributive, cardiogenic, hypovolemic, neurogenic, and obstructive, each with distinct causes and hemodynamic profiles.
- Initial treatment for most shock patients involves intravenous fluids to stabilize blood flow and organ perfusion, except in cases of pulmonary edema.
- Distributive shock is managed with vasopressors like norepinephrine, which cause vasoconstriction by stimulating adrenergic receptors.
- Cardiogenic shock is treated with inotropes such as dobutamine and milrinone, which increase cardiac contractility and may cause vasodilation.
- Obstructive shock is treated by addressing the underlying cause, such as thrombolytics for pulmonary embolism or surgical interventions for cardiac tamponade and tension pneumothorax.

Review Questions

1. A 65-year-old female presents with a 2-week history of cough, fever, and chills. She is confused and disoriented. Her blood pressure is 70/50 mm Hg, pulse is 125 beats/min, and temperature is 102.5°F. Exam shows right flank tenderness, and her urinalysis shows numerous white blood cells and bacteria. The white blood cell count is elevated. She is admitted to the intensive care unit. Her blood pressure does not change with 1 L of normal saline. Which of the following medications would be the best next step in management?

- A. Digoxin
- B. Dobutamine
- C. Milrinone
- D. Norepinephrine

Explanation (requires correct answer)

2. A 55-year-old male with a history of cardiomyopathy (heart failure with an ejection fraction of 10%) is admitted for consideration of heart transplant. He is found to be in cardiogenic shock with low cardiac output and high SVR. He is started on milrinone and listed for transplant. What is the mechanism of action of this medication?

- A. α_1 -Adrenergic stimulation
- B. β_1 -Adrenergic stimulation
- C. β_2 -Adrenergic stimulation
- D. PDE3 inhibition

Explanation (requires correct answer)

3. A 35-year-old female who is 16 weeks pregnant is admitted with confusion and cold extremities. Her blood pressure is 85/75 mm Hg, pulse 110 beats/min, and temperature is 100.4°F. The heart, lung, and

abdominal/pelvic exams are normal. Which of the following would be the most appropriate initial treatment?

- A. Give normal saline
- B. Start epinephrine
- C. Start milrinone
- D. Start norepinephrine
- E. Start vasopressin

Explanation (requires correct answer)

4. Which type of shock is caused by sympathetic output?

- A. Distributive shock
- B. Cardiogenic shock
- C. Hypovolemic shock
- D. Neurogenic shock
- E. Obstructive shock

Explanation (requires correct answer)

How would you rate this Brick?



References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 Describe the hemodynamic findings for each type of shock.

- 02** Explain the use of fluid resuscitation in shock.
- 03** Justify the use of pharmacological agents in shock treatment
- 04** Outline the treatment plan for each type of shock

Objective Confidence: 0/4